

Major Project Presentation (2026)

- **ID:** PCSE27-12
- **Project Title:** Detecting Plant Leaf Diseases with Swin Transformer V2-B For Sustainable Farming
- **Team Members:**

Sidhant Pandey	(2300290100252)
Shivam Arora	(2300290100232)
Sarthak Goel	(2300290100226)
Shudha Nidhi Soni	(2300290100250)



Problem Statement

Plant diseases represent a persistent threat to global food security. Current diagnostic techniques relied upon by farmers through physical observation are largely time-consuming, subjective, and inaccessible in underserved, off-site areas.

Time-Consuming & Subjective Diagnosis



Manual inspection by experts is slow and relies on individual judgment, leading to inconsistent results and delayed treatment decisions.

Limited Access in Rural Areas



Smallholder farmers in remote regions lack access to agricultural extension officers and pathologists for proper disease identification.

Integration and Deployment Gaps in Existing AI Solutions

Current diagnostic systems fail to combine visual image data with qualitative, multi-lingual farmer observations, lacking field-ready deployment accessibility and real-time epidemiological (How often diseases occur in different groups) outbreak tracking.

Solution Proposed

Technologies used:

- Frontend : React.js, React Native, HTML5 / CSS3
- Backend : Node.js, Express.js, REST API + Fast-API
- Database: MongoDB
- AI Model: Swin Transformer V2-B
- Deployment: Cloud-ready Infrastructure (AWS)

Why Vision Transformers (SwinV2-B)?

Traditional AI (CNNs)

- ✗ Looks at small pieces of the image one by one
- ✗ Struggles when disease is partially hidden
- ✗ Works well in labs, poorly in real farms

Vision Transformers (Better)

- ✓ Sees the entire leaf at once, like a human doctor
- ✓ Understands relationships between different parts
- ✓ Works on real farm photos with clutter & lighting

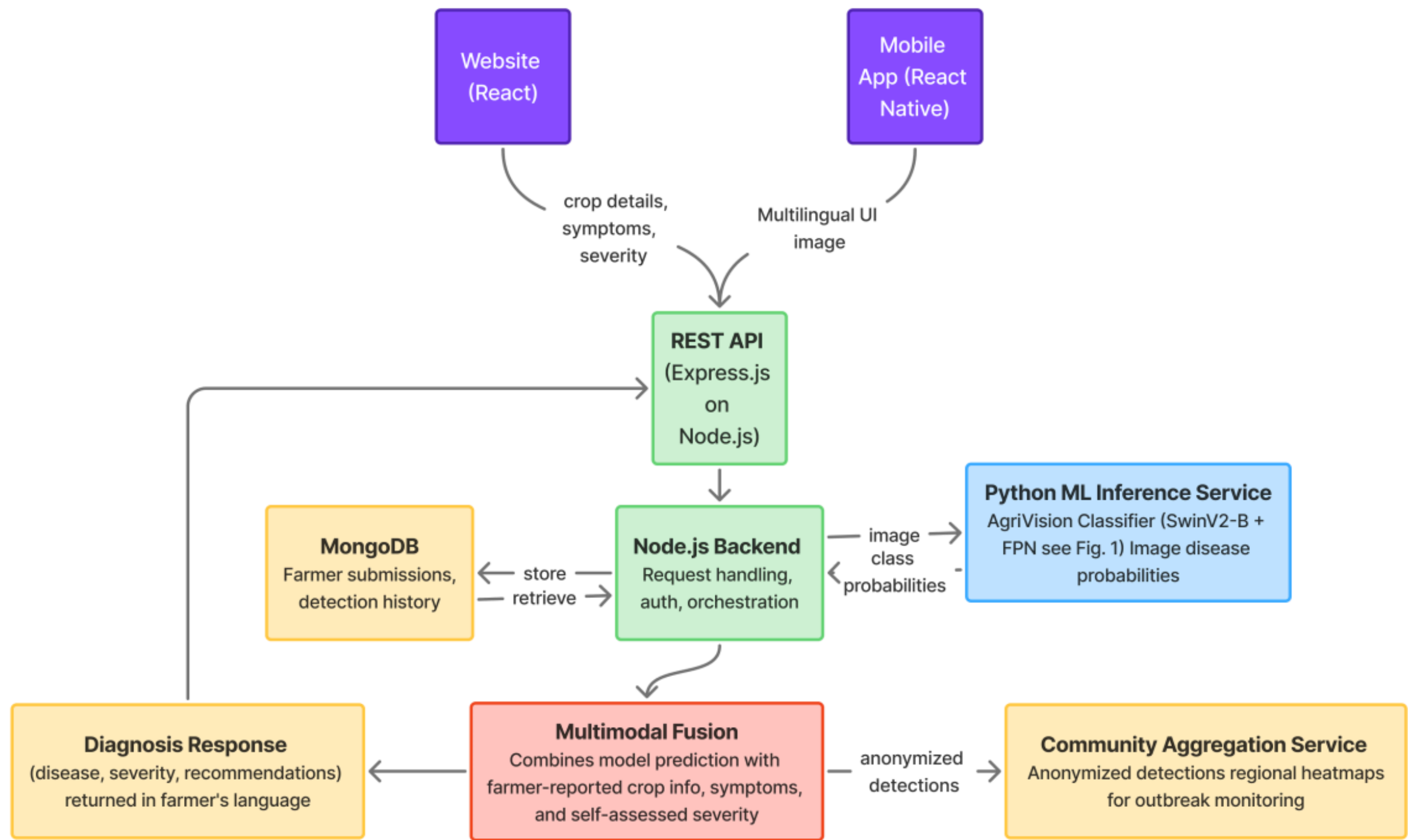


- **Intelligent Image Analysis:** Advanced AI model analyzes leaf photographs to instantly identify diseases with high accuracy, eliminating subjective assessment.
- **Context-Aware Diagnosis:** Combines AI predictions with farmer-provided information (symptoms, crop type, severity) for more reliable and localized results.

- **Real-Time Outbreak Heat-Mapping:** Integrates anonymized geospatial tracking to generate real-time epidemiological heatmaps, enabling early detection and localized containment of community-level crop disease outbreaks.
- **Multilingual & Mobile-First:** Web and mobile platform with multilingual support ensures accessibility for farmers worldwide, regardless of language or location.



Flow Diagram



AgriVision Platform: Overall System Architecture



RESEARCH & REFERENCES

Research Paper

- [1] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.
- [2] G. Geetharamani and A. Pandian, "Identification of plant leaf diseases using a nine-layer deep convolutional neural network," *Computers & Electrical Engineering*, vol. 76, pp. 323–338, 2019.

Research Paper

- [3] P. Mahapatra, M. Panda, S. K. Dash, and U. K. Sahu, "Advancing plant disease classification using an attention-based CNN for intra-dataset and cross-dataset training," *Scientific Reports*, vol. 16, p. 10925, 2026.
- [4] U. Mishra, A. Pandey, G. Logeswari, and K. Tamilarasi, "Deep learning- based disease detection in potato and mango leaves: A comparative study of CNN, AlexNet, ResNet, and EfficientNet," *Scientific Reports*, vol. 16, p. 2788, 2025

Research Paper

- [5] I. Pacal, I. Kunduracioglu, M. H. Alma, M. Deveci, S. Kadry, J. Nedoma, V. Slany, and R. Martinek, "A systematic review of deep learning techniques for plant diseases," *Artificial Intelligence Review*, vol. 57, p. 304, 2024.
- [6] Y. M. Alsakar et al., "Eff-swin-hgso: Attention-driven and optimized plant leaf disease diagnosis using EfficientNetV2B0 and Swin transformer with HGSO feature selection," *The Journal of Supercomputing*, 2025.
- [7] D. Singh, N. Jain, P. Jain, P. Kayal, S. Kumawat, and N. Batra, "PlantDoc: A dataset for visual plant disease detection," in *Proceedings of the 7th ACM IKDD CoDS and 25th COMAD*, 2020, pp. 249–253.

Transforming Agriculture with AI

*Bringing precision diagnosis
and sustainable farming to farmers worldwide*

